



Large-scale reduction of ungulate browsing by managed sport hunting

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ABSTRACT

Browsing by free-living ungulates is a serious threat to forest regeneration. Although predictors of browsing and potential consequences are well researched, less information is available in the scientific literature about possible solutions to the problem. We investigated whether a large-scale game management system in a traditional sport hunting culture has the potential to effectively reduce browsing damage. Our analysis was based on browsing survey data from 2006 and 2009 covering more than 25,000 km² in Bavaria, Germany. After an initial assessment of browsing damage throughout Bavaria in 2006, game management plans were altered, and the effect of this intervention was assessed in 2009. Browsing damage clearly declined in 2009 in areas where the suggested deer harvests were increased in 2006 game management plans. This is the first report of a successful implementation of a large-scale game management system aiming at reducing browsing damage to facilitate unfenced forest regeneration.

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1. Introduction

Ungulates historically formed complex communities of grazers and browsers in Europe. The ungulate-vegetation interactions during the last 500,000 years have been primarily driven by climatic variation of the glacial–interglacial cycle (Bradshaw et al., 2003). Climate change and human activities during the most recent glacial cycle have accelerated extinction rates of several ungulate species. Only a reduced community of small-to-medium-sized free-living herbivores has survived modern man in most temperate zones of Europe. Large grazers subsist only as livestock (Mayer and Huovinen, 2007). Populations of the few remaining browser species in large parts of North America and Europe, particularly roe deer *Capreolus capreolus* L. in Europe and White-tailed Deer *Odocoileus virginianus* L. in the United States, have increased over the last century and are still increasing (Milner et al., 2006). Also the distribution ranges of roe deer are increasing in various Central European countries (Andersen et al., 1998; Kuijper et al., 2009), such as Great Britain (Ward, 2005) and Spain (Tellería and Virgós, 1997) because of, for example, protection from overhunting, selective harvesting regimes with the aim to increase the number of trophy bucks, and supplementary feeding (Milner et al., 2006).

The impact of ungulates on the landscape and forests is still a matter of debate (Svenning, 2002). High ungulate densities

and there from extensive browsing pressure can have contrasting effects on the biodiversity of fauna and flora (Stewart, 2001; Feber et al., 2001). Positive effects include an increase in dung beetle abundance and forest gaps being kept open for many thermophilic insects in forests (Putman, 1996; Partl et al., 2002). Negative effects include a reduction in the host plants of rare insects and lower regeneration of several tree species, including many of economical importance (Stewart, 2001). This suppression of regeneration of specific tree species through selective browsing can lead to a significant decrease in their proportions in future forest generations and has been identified as one of the major problems forest managers encounter when aiming at converting coniferous plantations to near-natural mixed-forests stands (Palmer and Truscott, 2003a). Quantitative studies have shown that browsing directly affects the growth, reproduction, and survival of seeds and saplings and has the potential to lead to extinction of palatable species. Rooney et al. (2002) concluded that deer reduce Northern White Cedar *Thuja occidentalis* L. regeneration, and Odermatt (1999) reported that fir *Abies alba* L. regeneration in the Swiss Alps was eliminated by browsing (similar results were reported by Ammer, 1996, for the Bavarian Alps). Advanced forest regeneration is particularly sensitive to deer presence (Gill and Morgan, 2010). The multitude of effects of browsing by ungulates in high densities on forest regeneration has been reviewed by Gill (1992) and Côté et al. (2004).

The problem of ungulate browsing was first brought to the attention of a large scientific audience by Leopold (1936a,b) with a description of the deer–Dauerwald (“permanent forest”) dilemma in pre-World War II Germany. He recognized that high densities of ungulate populations in German commercial coniferous forest plantations rendered natural forest regeneration impossible. The

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description of Leopold (1936a,b) remained the *status quo* in German forests until the 1980s, when air pollution and later climate change attracted public interest in forests and their tree species composition and resulted in legislation to permit implementation of a near-natural forest system. A transition from coniferous plantations towards mixed forests – or, in other words, toward a Dauerwald – began (see Schabel, 2001, for a more-detailed overview of these developments).

To facilitate this alteration of silvicultural systems toward near-natural forests forestry in Bavaria, both the Bavarian Forest and the Bavarian hunting laws aimed at maintaining free-living ungulates at population densities that allow natural regeneration of locally abundant tree species without artificial protection such as fencing. To support this state-wide plan, an accompanying large-scale quantitative monitoring of ungulate browsing was implemented in the Bavarian game management system in 1986 (Kennel, 1988). The Bavarian hunting law requires a detailed management plan for Roe Deer, red deer *Cervus elaphus* L., and Chamois *Rupicapra rupicapra* L. for each local hunting district (usually between 0.8 km² and 10 km²), specifying the number of males, females, and fawns to be harvested in a 3-year period. An assessment of ungulate browsing was introduced in 1986; since then, the management has been planned based on the locally estimated proportion of browsed trees and aims at reducing browsing through hunting. The procedure is costly (approx. 2 million € for each survey) and time consuming. Its efficacy and usefulness has been continuously questioned throughout the last 25 years (for example Meidel, 2008), but no quantitative measures have been considered so far.

The general efficiency of hunting to regulate animal populations varies considerably among scales from local forests to state-wide actions and also among species. For ungulates in temperate forests, serious doubt whether sport hunting can be effective in reducing browsing damage has been voiced in the literature. For example, Tanentzap et al. (2009) observed only a limited change in the vegetation in New Zealand even after a drastic reduction in red deer since 1964; the plant recovery was slow and incomplete after nearly 40 years of game management. Kamler et al. (2010) have shown that reduced deer densities lead to reduced browsing on spruce and other non-palatable tree species, but not on deciduous trees, which are the most important tree species for the conversion to near-natural forests. A similar result was reported by Månsson (2009), who studied moose *Alces alces* L. densities and browsing damage in Sweden. Kuijper et al. (2009) argue that deer feed in forests gaps, and thus browsing is concentrated in these areas, which leads to a highly nonlinear relationship between deer densities and browsing. They concluded that a reduction in browsing would require a drastic reduction in population sizes, which, according to Kuijper et al. (2009), would hardly be feasible. Contrary to these findings, Tremblay et al. (2006) and Tremblay et al. (2007) observed an exponential increase in forest regeneration indicators when deer densities declined over 3 years. Successful regeneration after a substantial reduction of deer densities was reported by Behrend et al. (1970) and Gaston et al. (2008). This disparity between the studies might partially be due to the small study areas and small sample sizes, and the large differences in forest cover, species composition, and climate on the landscape scale. In addition, the density-dependent response of the deer population, the density of herbivores and the density of saplings interact in forest ecosystems (Morellet et al., 2007). These and other factors may affect the deer density as well as the proportion of browsing damage.

Despite the high relevance of browsing from both an economical and a conservation point of view, we are aware only of one large-scale study addressing the efficacy of hunting to reduce browsing damage: Hörnberg (2001) showed a decline in browsing damage in Swedish forest inventories with decreasing moose population lev-

els estimated from moose harvest data. Here we address whether large-scale game management based on a quantitative assessment of forest regeneration is effective in reducing browsing damage. We specifically tested the hypothesis that interventions in game management plans, such as a proposed increase in harvest numbers, leads to a measurable rapid reduction in ungulate browsing. We based our study on official survey statistics, including data of ca. 3.2 million plants surveyed in 2006 and 2009 in a study area of approximately 25,000 km² in Bavaria, Germany.

2. Materials and methods

2.1. Survey area

Bavaria is the largest federal state of Germany, located in the southeastern part of the country (48° 46' N 11° 25' E). With an area of 70,500 km², it covers almost 20% of the total land area of Germany. More than one-third (36%) of the area, 25,000 km², is covered by forests, 32% of which are state owned; thus, most forests are in private hands. The number of hunters is estimated at 51,000, that is, 0.41% of the population of almost 12.5 million.¹ Hunting rights are held by the landowner but are mostly leased to paying leaseholders, except for some large privately owned forests or parts of the state forests, where hunting is amongst the duties of professional hunters or forest officers. Even there, more than 50% of the harvests are conducted by non-professional guest hunters. Therefore, the vast majority of harvests in Bavaria are conducted by sport hunters.

Roe deer are abundant and distributed throughout Bavaria. Annual harvest numbers are still increasing (1982/83: 218,226; 2008/09: 277,555, Deutscher Jagdschutzverband, 2010). In contrast, red deer are no longer found outside of red deer management areas, in which the harvest numbers average around 10,700 animals per year (Deutscher Jagdschutzverband, 2010). Chamois are restricted to the Alps, with approximately 4400 individuals taken every year (Deutscher Jagdschutzverband, 2010). Other large herbivores do not account for similarly large numbers and are therefore not considered in our study.

2.2. Game management system

Bavaria is subdivided into 762 game management districts (in German “Hegegemeinschaften”, see Fig. 1), an association of all local hunting districts and their owners and leaseholders. The primary aim of this association is the large-scale management of ungulates by means of coordinating game management plans for all associated local hunting districts. The actual number of animals taken in each hunting district must be reported to the local hunting authority, but the numbers are not publicly available.

2.3. Browsing survey

The amount of ungulate browsing is monitored on a regular basis by the Bavarian Forest Administration (Forstliches Gutachten, 2006, 2009). This survey covers all 25,000 km² of forests in Bavaria regardless of ownership and is conducted every 3 years. It was first established in 1986, and the last survey took place in 2009. The administrative organization of forest and hunting authorities was significantly altered in 2005, rendering a direct comparison with earlier survey results difficult. Therefore, we restrict our analyses to data gathered during the two most-recent surveys in 2006 and 2009.

¹ <http://www.jagd-bayern.eu/Praxis-der-Jagd-in-Bayern.116.0.html>.

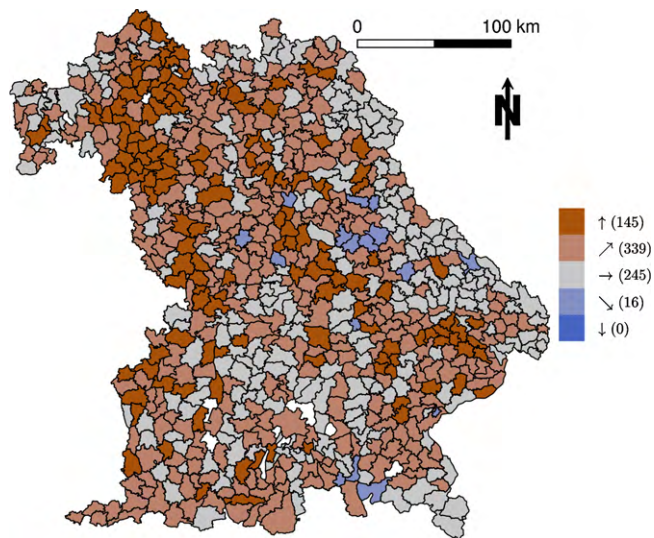


Fig. 1. Game management districts in Bavaria with color-coded interventions (based on 2006 survey) on harvest number: substantially decrease (↓), decrease (↘), no change (→), increase (↗), and substantially increase (↑). Total numbers of game management districts in these groups are given in parentheses. White areas: no information available.

The survey is based on a fixed regular grid that defines sampling points at the intersections. In both 2006 and 2009, at each intersection, the nearest appropriate sampling area to measure browsing proportions was inspected, i.e., a forest area with tree regeneration of a sufficient density (1300 trees per hectare) and size (the area must accommodate a transect of 50–100 m). In each relevant regeneration area, 75 saplings between 20 and 130 cm height, i.e., established saplings up to the maximal height reached by Roe Deer, located on five plots, corresponding to 15 saplings per plot, along the transect were examined for browsing on leading shoots. The proportion of saplings suffering from leading shoot browsing and the expected damage is strongly correlated (Odermatt, 1999; Palmer and Truscott, 2003a; Didion et al., 2009), and leading shoot browsing and browsing in general are dependent (Welch et al., 1991). The proportion of saplings showing leading shoot browsing is a measure commonly used to assess browsing (Rooney and Waller, 2003; Gill, 1992; Palmer and Truscott, 2003a; Kamler et al., 2010; Gill and Morgan, 2010, among others). The saplings were classified according to genera into nine classes: spruce *Picea*, fir *Abies*, oak *Quercus*, beech *Fagus*, pine *Pinus*, larch *Larix*, ash/maple/elm/linden *Fraxinus/Acer/Ulmus/Tilia*, other softwoods, and other hardwoods (rowan berry *Sorbus aucuparia*, birch *Betula*, alder *Alnus*, hornbeam *Carpinus betulus*, willow *Salix*, poplar *Populus*, and service tree *Sorbus domestica*). The sampling scheme is described in more detail in Kennel (1989) and the official survey reports (Forstliches Gutachten, 2006, 2009). The main outcome of the two surveys was the proportion of saplings within each of the nine classes suffering from ungulate browsing, reported separately for each of the game management districts. For 2006, information on 745 game management districts was available for our analysis; for 2009, the results are based on data for 759 game management districts.

2.4. Interventions in game management plans

Based on the assessment of browsing obtained from the 2006 survey, the local forestry authorities issued the recommendations for game management between April 2007 and March 2010 to the respective game management districts and hunting authorities. The recommendation was standardized on a five-point scale:

decrease harvest numbers substantially (↓), decrease harvest numbers (↘), do not change harvest numbers (→), increase harvest numbers (↗), and increase harvest numbers substantially (↑), all relative to the harvest numbers implemented in 2003. Each recommendation was made at the scale of the game management district, but no assessments or recommendations for single local hunting districts are available. The practical implementation of these suggestions for each local hunting district was then negotiated between the landowner and leaseholder of the local hunting district, the game management district (most important between neighboring hunters), and the local hunting authority in a democratic discussion process. Detailed data on these management plans was, however, not available. Thus, the recommendations made for each of the 745 game management districts with data available in 2006 was used as a surrogate for the actually implemented management action, and we refer to this variable as “suggested intervention” in the following. The statistical analysis is subdivided into two parts: reliability analysis and efficacy analysis.

2.5. Reliability analysis

We tested the reliability of the suggested intervention in game management made by local forestry authorities. For the whole procedure to be meaningful and acceptable, the intervention must reliably depend on the locally estimated browsing damage. Based on interventions suggested for the game management districts in 2006, we fitted a regression model that took the proportion of browsed saplings for each of the nine tree genus classes as explanatory variables. To allow for a more-complex statistical description of the process of making a suggestion for an intervention based on browsing proportions, we fit a conditional regression tree model (Hothorn et al., 2006b) and a conditional random forest model (Strobl et al., 2007) to data obtained in the 2006 survey, with each game management being one observational unit. Such tree models explain variation in categorical response variables (intervention) as a function of the explanatory variables (browsing proportions). One explanatory variable is selected from all available exploratory variables by maximizing the association with the response via a permutation test (Hothorn et al., 2006a), and a split point that separates the response values into two homogeneous groups is estimated by a maximally selected test statistic (Müller and Hothorn, 2004; Hothorn and Zeileis, 2008). The split point is determined by a numerical value of the explanatory variable. Once the split point has been estimated for a selected explanatory variable and the groups have been defined, each group is further separated with new explanatory variables and split points. A stopping criterion based on permutation tests is used to stop the recursive splitting procedure (Hothorn et al., 2006b). A random forest (Breiman, 2001) is an average of 500 randomized trees fitted to bootstrap samples of the 2006 game management data. This method is one of the best contemporary off-the-shelf classifiers.

We tested the null hypothesis that interventions are independent of survey results, i.e., that the suggestions for interventions are made arbitrarily or subjectively by local forestry authorities, using a linear-by-linear association test (Agresti, 2002) of actual interventions suggested in 2009 and the predictions of the two models based on 2009 browsing proportions. This first part of the analysis used publicly available data from the 2006 and 2009 surveys (Forstliches Gutachten, 2006, 2009).

2.6. Efficacy analysis

We measured the efficacy of this intervention by means of the odds-ratio of ungulate browsing between 2006 and 2009, separately for the nine tree genus classes and the five potential

Table 1

Overview of ungulate browsing surveys in Bavaria in 2006 and 2009. A total number of 3,241,635 trees were inspected for leading shoot browsing.

Tree genus class	Year	Trees browsed	Trees surveyed	% Trees browsed
Spruce	2006	51,706	771,720	7%
	2009	30,395	702,714	4%
Beech	2006	56,583	296,042	19%
	2009	47,144	310,801	15%
Ash/Maple/Elm/Linden	2006	82,559	238,903	35%
	2009	76,149	263,514	29%
Other hardwoods	2006	47,493	140,130	34%
	2009	47,107	153,355	31%
Oak	2006	26,510	74,111	36%
	2009	26,348	81,099	32%
Fir	2006	14,278	50,402	28%
	2009	11,309	56,753	20%
Pine	2006	2402	45,471	5%
	2009	2000	42,689	5%
Other softwoods	2006	633	3930	16%
	2009	1045	5621	19%
Larch	2006	178	1841	10%
	2009	263	2539	10%
Total	2006		1,622,550	
	2009		1,619,085	

interventions:

$$OR_{g,r} = \frac{\mathbb{P}(\text{browsing}|g, r, 2009)/\mathbb{P}(\text{no browsing}|g, r, 2009)}{\mathbb{P}(\text{browsing}|g, r, 2006)/\mathbb{P}(\text{no browsing}|g, r, 2006)},$$

where g is one of the nine tree genus classes and r is one of the five potential interventions. The odds of browsing damage in 2006 for genus class g and intervention r is given in terms of the ratio of the conditional probabilities for browsing and no browsing, $\mathbb{P}(\text{browsing}|g, r, 2006)/\mathbb{P}(\text{no browsing}|g, r, 2006)$, and analogously for 2009. Values of $OR_{g,r} < 1$ indicate a decrease of browsing between 2006 and 2009 whereas values > 1 indicate more severe browsing in 2009 compared to 2006. The null hypothesis states that all $9 \times 5 = 45$ odds-ratios are $= 1$:

$$H_0 : OR_{g,r} = 1 \text{ for all genera } g \text{ and } r \in \{\downarrow, \searrow, \rightarrow, \nearrow, \uparrow\},$$

i.e., that the intervention in 2006 showed no effect at all on browsing in 2009. We were especially interested in monotone alternatives, that is, in patterns where the odds-ratios for browsing decrease with increased recommended harvest number.

Treating each single tree inspected in either 2006 or 2009 as an observational unit, we estimated the odds-ratios by means of a logistic linear mixed model (Pinheiro and Bates, 2000). To address potential spatial heterogeneity present in the data, we included a random intercept into the model for each game management district. Odds-ratios were estimated by the exponential transformation of regression coefficients defined by the interaction of genera and intervention for trees inspected in 2009, thus treating 2006 as the baseline. Simultaneous 95% confidence intervals for all estimated odds-ratios were computed based on the methodology described by Hothorn et al. (2008).

This analysis used publicly available data of the 2006 and 2009 surveys (Forstliches Gutachten, 2006, 2009) and the total numbers of saplings inspected for each of the nine tree classes in each game management district, made available by the Bavarian State Ministry of Agriculture and Forestry (personal communication).

2.7. Computational details

These analyses were performed using R version 2.11.0 (R Development Core Team, 2010). Conditional inference trees and conditional inference forests were fitted using the R add-on package **party** (Hothorn et al., 2010b). The logistic linear mixed-effects model was fitted using package **lme4** (Bates and Maechler, 2010); simultaneous confidence intervals for odds-ratios were obtained

from functionality implemented in package **multcomp** (Hothorn et al., 2010a).

3. Results

In total, 3,241,635 saplings on approximately 25,000 km² of forest in 745 (2006 survey) and 759 (2009 survey) game management districts were inspected for leading shoot damage caused by browsing. The relative and total numbers of saplings suffering from ungulate browsing in 2006 and 2009 are given in Table 1. Overall, the proportion of saplings with browsed leading shoots appeared to decrease between the 2006 survey and the 2009 survey for most tree genus classes (Table 1). The interventions in game management plans made in 2006 are color-coded in Fig. 1. No game management district was advised to reduce harvest numbers substantially; most were advised to increase the number of animals harvested (Fig. 1).

3.1. Reliability analysis

The conditional inference tree based on data gathered in only 2006 (Fig. 2) clearly links higher browsing proportions to interventions of increased harvest numbers. Using the data of 2009 as an independent validation sample, we compared the observed interventions (suggested by forestry authorities based on 2009 browsing data) with the predicted interventions obtained from the two regression models (conditional inference tree model and random forest model, Table 2). The models fitted only on 2006 data accurately predict 2009 interventions. The off-diagonal numbers are incorrect predictions. Note that only 21 (conditional inference tree) or 13 game management districts (conditional inference forest) are severely incorrect (no change vs. substantial increase or vice versa). The prediction performance (Table 2) of conditional random forests is only slightly better than the prediction performance of the conditional inference tree (Fig. 2). Thus, the model in Fig. 2 is almost as good as the much more complicated random forest model, and we can draw conclusions from the simpler conditional inference tree model without much loss of information. The conditional inference tree indicates that the proportion of leading shoot browsing of other hardwoods (rowan berry, birch, alder, hornbeam, willow, poplar, or service tree) is highly discriminant, followed by the widely distributed tree species beech; at higher levels of browsing impact, spruce is a good predictor for this representative browsing model. The model depicted in Fig. 2 reflects the

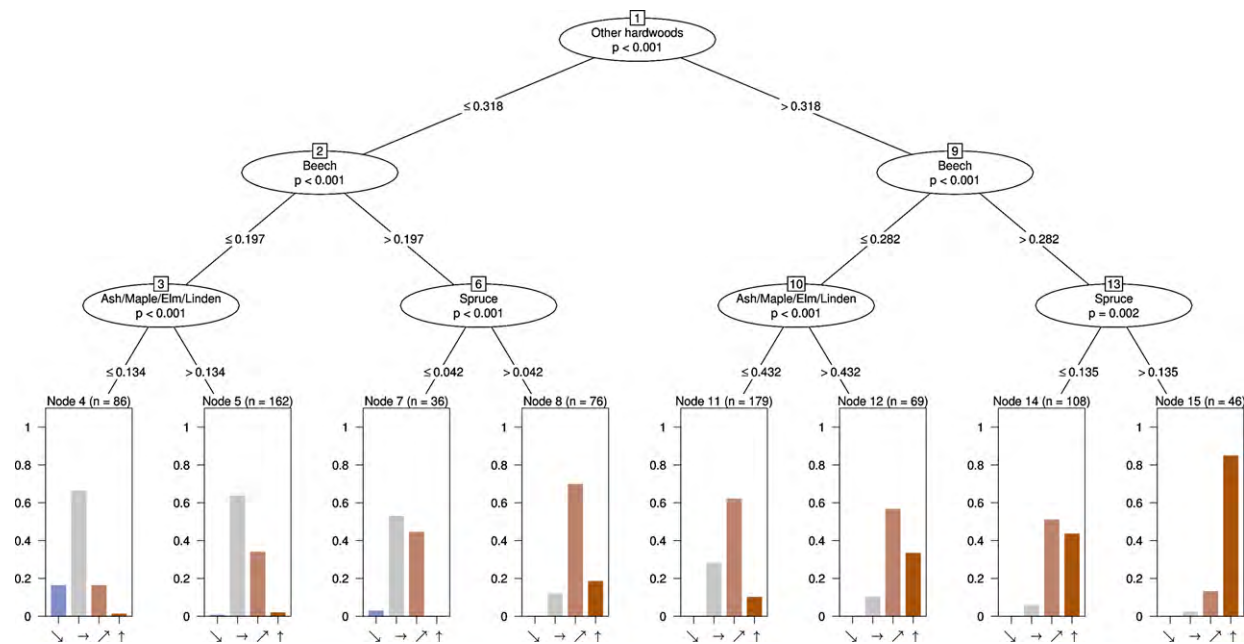


Fig. 2. Reliability analysis. Conditional inference tree model linking browsing proportions for the different tree genus classes to the suggested interventions in 745 game management districts in 2006. The barplots depict the distribution of 2006 interventions, i.e., decrease (↓), no change (→), increase (↗), and substantially increase (↘) harvest numbers, for the subgroups of game management districts defined by cut-offs in browsing proportions for certain tree genus classes. *P*-values correspond to multiplicity-adjusted permutations tests of independence between browsing proportions and intervention. The model can be interpreted as a representation of the decision process. Other hardwoods seems most important for the assessment of browsing: when <31.8% of these saplings showed leader browsing, and beech and ash/maple/elm/linden browsing was low (<19.7% and <13.4%, respectively), the majority of game managements were advised to maintain or reduce harvest numbers.

decision-making process of the forest authorities, averaged over all game management districts. In general, game management districts with a browsing proportion of other hardwood species >32% were never advised to reduce harvest numbers. If, in this group of districts, also >28% of beech saplings and 13% of spruce saplings showed signs of browsing on leading shoots, the suggestion for intervention was to substantially increase harvest numbers for the majority of these game management districts (39 out of 46, Fig. 2, node 15).

Formally, the null hypothesis of independence of model predictions and actually observed interventions (see Table 2) can be rejected at nominal level $\alpha = 0.05$ for both the conditional inference tree model ($\chi^2 = 139.41$ at one degree of freedom) and conditional inference forests ($\chi^2 = 209.846$ at one degree of freedom). Thus, the suggested interventions for game management plans by forestry

authorities are reliably based on proportions of saplings suffering from ungulate browsing and are thus a sound basis for game management.

3.2. Efficacy analysis

We could estimate only 36 of 45 odds-ratios because no suggestions for substantially reduced harvest numbers were made. The corresponding parameters were therefore omitted. Point estimates and simultaneous confidence intervals for browsing odds-ratios between 2006 and 2009 are depicted in Fig. 3. This plot clearly shows that increased and substantially increased harvest numbers are associated with a reduced browsing odds for spruce, beech, ash/maple/elm/linden, other hardwoods, oak, and fir. For pine, the confidence interval for substantially increased harvest numbers contains 1; therefore, no effect of higher suggested harvest numbers is detectable, but the number of observations in this category is rather low. None of the confidence intervals excludes 1 for other softwoods and larch; therefore, no effect of the intervention of browsing on these species can be postulated for these small numbers of trees. In summary, the administrative suggestion to increase harvest numbers starting in 2006 resulted in reduced ungulate browsing proportions after a 3 year period, until 2009, for the most important tree species or genera.

For areas with recommendation not to change harvest numbers, browsing was reduced only for spruce and fir, and remained unchanged for beech and ash/maple/elm/linden. Remarkably, keeping the number of animals harvested constant led to increased browsing of other hardwoods and pine. Although only 16 game management districts were advised to reduce the number of animals harvested and thus only a small number of trees correspond to this category, browsing on all tree species other than spruce increased substantially. Therefore, unchanged or decreased harvesting have lead to increased browsing for some tree species within 3 years.

Table 2
Reliability analysis. Predicted vs. observed interventions based on regression models linking browsing proportions to interventions. (A) Conditional inference tree fitted to 2006 data only (the model is depicted in Fig. 2). (B) Conditional random forest fitted to 2006 data only. Diagonal elements indicate correct predictions, off-diagonal elements correspond to erroneous predictions. See text and Fig. 1 for an explanation of the arrows.

Observed 2009	Predicted 2009				
	↓	↘	→	↗	↑
(A) Conditional inference tree					
↓	.	.	.	.	.
↘	.	.	7	.	.
→	.	.	252	94	7
↗	.	.	110	181	14
↑	.	.	14	66	14
(B) Conditional inference forest					
↓	.	.	.	.	.
↘	.	.	7	.	.
→	.	.	247	97	9
↗	.	.	82	204	19
↑	.	.	4	62	28

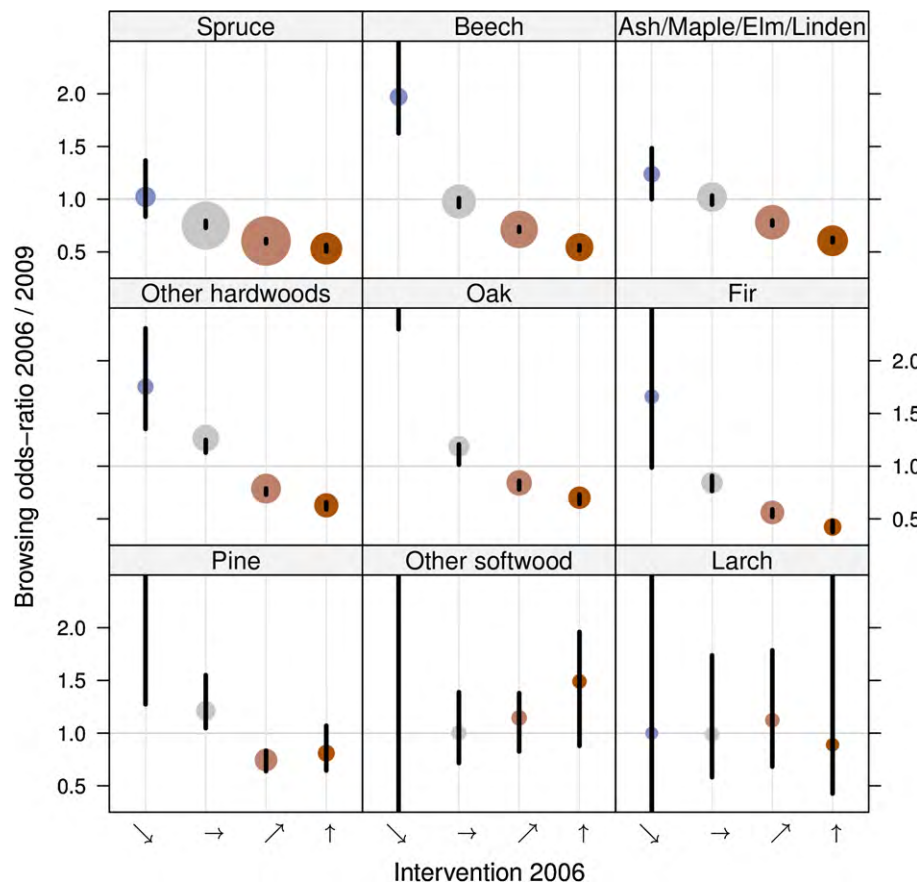


Fig. 3. Efficacy analysis. The estimated browsing odds-ratio (based on $n = 3,241,635$ observations) between 2006 and 2009 for each combination of tree genus class and 2006 intervention at levels decrease ($\searrow$), no change ($\rightarrow$), increase ($\nearrow$), and substantially increase ($\uparrow$) harvest numbers. An odds-ratio of 1 means no change in browsing proportions between 2006 and 2009, >1 an increased browsing and <1 reduced browsing. Estimates are given as color-coded circles, whose radius is proportional to the square-root of the number of trees of the corresponding class sampled. Black lines denote simultaneous 95% confidence intervals for all 36 estimated odds-ratios. Note that values >2.5 are omitted from this plot.

4. Discussion

Ecological and economical aspects of ungulate browsing, mainly by roe deer, with its strong impact on tree regeneration and species composition (Didion et al., 2009), triggered the development of management concepts, that are mainly based on the assessment of browsing damage. Our main finding is that ungulate browsing can be reduced substantially at a state-wide scale within a very short period of 3 years, even with a hunting system dominated by sport hunting. To the best of our knowledge, a similar finding on such a large scale has never been published, and, in light of previously published results on the effect of sport hunting on ungulate browsing, our results are remarkably clear. The odds of browsing on spruce, beech, and fir saplings were reduced as much as 50% in 3 years between 2006 and 2009. Although only summary statistics from browsing surveys in the 1980s are available, the browsing proportions clearly continuously declined. Kennel (1988) reported 31% browsing on spruce, 47% on fir, 43% on beech, and 54% on other hardwoods. The figures given in Table 1 and the statistical analysis of the differences between 2006 and 2009 indicate that interventions in game management plans based on an assessment of browsing damage are an appropriate and efficient way to reduce browsing substantially. However, the results should not be misinterpreted as meaning that browsing declined in *all* game management districts; the estimated numbers are, in the sense of a logistic regression model, averages over game management districts.

The impact of large ungulates on tree regeneration was monitored and assessed according to the proportion of saplings suffering from leading shoot browsing. Although there is a strong connection between ungulate densities and browsing (e.g., Rooney et al., 2002; Tremblay et al., 2006, 2007; Kamler et al., 2010; Gill and Morgan, 2010), Palmer and Truscott (2003b) and Palmer and Truscott (2003a) pointed out that successful regeneration cannot be predicted from deer densities alone. Data on ungulate densities, such as the actual number of deer present in each of the game management districts, were not available for our analysis, and thus our results do not directly allow conclusions about changes in ungulate densities. Several other predictors may influence the proportion of browsing damage, such as the proportion of forest, fragmentation, climate, and soil nutrients on the landscape scale. Our understanding of the complex set of variables that determines the level of ungulate populations in heterogeneous cultivated landscapes is still restricted, and further analyses are required.

Furthermore, because of the lack of harvesting data for the hunting districts, the observed effect cannot simply be attributed to reduced ungulate densities. The effect of hunting on tree regeneration might be caused by changes in animal behavior induced by increased hunting pressure rather than the actual harvest number (Martin and Baltzinger, 2002). Also, forest management might have increased the availability of additional food for ungulates, which might lead to reduced browsing (Miller et al., 2009). Thus, the effect might be achieved at ungulate densities remaining at the same high level, for example, by altered hunting strategies, such

as placing more hunting pressure near regeneration areas. Such alterations in hunting behavior and strategies may very well affect small studies conducted in restricted areas. However, for our large-scale investigation, it seems quite unrealistic that such a broad alteration of hunting strategies can be initiated through state-wide hunting management plans. Therefore, it seems more reasonable that reduced browsing is a consequence of reduced densities of ungulates in forests.

The proportion of sufficient forest regeneration is only an indirect measurement of sufficient forest regeneration, the major aim of forest managers. However, this surrogate can probably be reliably used because the proportion of saplings suffering from leading shoot browsing and successful forest regeneration are strongly correlated (Odermatt, 1999; Palmer and Truscott, 2003a; Didion et al., 2009) to forest regeneration. However, one problem of the survey design is that browsed saplings <20 cm in height were not part of the assessment. This can lead to an underestimation of browsing impact on rare and palatable species. Interestingly, forest authorities seem to highly value these palatable species, even though most of them are not of economic value. For example, in Fig. 2, the strongest predictor for interventions towards higher harvest numbers is the proportion of other hardwood species (e.g., rowan berry and alder). Even spruce, known as a non-palatable species, is in general an indicator of high browsing damage. Consequently, spruce browsing >13.5% almost always leads to a recommendation to substantially increase the harvest (Fig. 2, node 15). In addition, spruce browsing at lower levels (>4.2%) is associated with the suggestion to increase the harvest (Fig. 2, node 6). Thus, considering a broader set of trees or species aggregates provides a more robust system of assessment on several levels and can help to implement an early warning system (Rooney and Waller, 2003). Early warning systems are desirable since an observed increased deer browsing indicates that the actual damage is already underway.

The results reported here allow an assessment of the efficacy of managed sport hunting only on the state level. An assessment of individual effects, for example changes in browsing proportions for each game management district, would be valuable for local forest and hunting authorities. Also, the evaluation of predictors for local changes is an interesting research question: forest coverage or the proportion of state-owned forests within the game management districts or socio-political variables might reveal patterns that explain the success or failure of game management on a local scale. Spatio-temporal regression models based on Markov random fields have the potential to address these future research questions.

Furthermore, to get a deeper ecological understanding of the hunter-herbivore-plant system, a more comprehensive analysis would be required, including information about ecosystem productivity (Elmhagen et al., 2010), browsing (Morellet et al., 2001), and deer density estimates (Morellet et al., 2007). Several indicators of ecological change in roe deer ecosystems on the level of population abundance, performance of individuals, and habitat impact have been developed for France (Morellet et al., 2007) and should be considered in addition to an assessment of browsing level for gaining a deeper understanding of forest ecosystems.

Whether sport hunting suffices to prevent deer overpopulation has been questioned. Côté et al. (2004) argue against its adequacy, mostly because they believe that hunters usually do not fully understand the effects deer have on ecosystems. Also Fuller and Gill (2001) were skeptical and point out that the scale of the deer browsing problem needs to be accepted by landowners, stalkers, and the public. Our results indicate that sport hunters could be motivated to comply with the goal of maintaining ungulates at appropriate densities by clearly defining management goals, recommending interventions based on a scientifically sound

assessment of ungulate browsing, and democratically discussing detailed game management plans on a local scale. The effect on forest regeneration is measurable and relevant.

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